BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 089 · Lesson 089 · The Processor City: Architecture Overview

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is the function of a program counter?

Q2. What does the ALU perform?

Q3. Name one type of bus.

Q4. A D flip-flop has D=1 and Q=0 just before the active rising edge. What is Q after the edge?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

ALU — Arithmetic Logic Unit.

CU — Control Unit.

register — Small fast CPU state storage.

bus — Shared communication pathway.

New learning

Observe — Inside CPU City Hall, the Control Unit organises operations, the ALU carries out arithmetic and logic, and registers hold immediate working state.

Explain — The Memory Warehouse stores program instructions and data. Address, data and control buses connect processor and memory; they carry different kinds of information.

Evaluate — Von Neumann architecture stores instructions and data in a shared memory/address space, whereas Harvard uses separate instruction and data memories/pathways.

Worked example

Explain a memory read using the city analogy and exact bus names.

1. Model response — The address identifies where; data is what is transferred. CU coordinates the process.

We Do · Practise with guidance

Explain why ALU and CU must be shown as separate parts of City Hall.

Model response

Saying the CU performs arithmetic confuses separate roles.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Name and describe ALU, Control Unit, and registers in processor. [6]

Question 2

Using correct names, explain address, data and control bus during memory read. [6]

Question 3

Evaluate usefulness and limitation of Processor City analogy for Von Neumann architecture. [6]

Exit Ticket · Answer before checking

The data bus carries the address of a memory location.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Holds address of next instruction.

Q2. Arithmetic and logical operations.

Q3. Address, data or control bus.

Q4. Q becomes1, as D is sampled at the active edge.

You Do pages · Independent questions

Q1. ALU arithmetic/logic [2], CU manages decoding/control [2], registers fast working storage [2].

Q2. Address bus carries requested memory location [2], control bus read request [2], data bus carries returned instruction/data bits [2].

Q3. Supports separate roles/paths [2], names correctly map [2], warns shared memory/bus contention and simplification of complex physical hardware [2].

Exit Ticket · Model answer

False. Address bus carries the location; data bus carries contents.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: